

1 Critical evaluation of the modified Gompertz response
2 function as a parametric template for
3 treatment-response trajectories in N-of-1 trial
4 simulation

5 pmsimstats team

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35 1 Abstract

36 **Background.** Simulation-based power analyses for aggregated N-of-1 trials re-
 37 quire a parametric template for the within- participant response trajectory. The
 38 pmsimstats framework adopts a modified Gompertz function for each of three
 39 latent response components (biological response, placebo-belief response, time-
 40 variant natural-history component). The choice of family is loaded into the simu-
 41 lated data before any analysis-model question is asked, so the implications of the
 42 family for power, type I error, and bias are foundational rather than peripheral.

43 **Methods.** We evaluate the modified Gompertz family against the relevant biolog-
 44 ical, pharmacokinetic, and biostatistical literature. We characterise the parameter
 45 space the function covers, the trajectories it can and cannot represent, and the
 46 sensitivity of standard pmsimstats inference to misspecification of the response-
 47 curve family. Three alternative families are compared as sensitivity analyses:
 48 the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and
 49 a piecewise-linear breakpoint model (no smoothness assumption). For each al-
 50 ternative we compute power, type I error, and bias of the biomarker-treatment
 51 interaction estimate at trial-relevant sample sizes against simulated data drawn
 52 from the alternative family but analysed under the standard linear-mixed model.

53 **Results.** A 1{,}000-replicate-per-cell simulation across 16 cells (4 trajectory fami-
 54 lies $\times$ 2 DGP architectures $\times$ 2 biomarker effect-size levels) at $N = 35$, OL+BDC
 55 design, $t_{1/2} = 0$, found trajectory family essentially irrelevant to power under
 56 both architectures at MCSE 0.013 to 0.016. Under Architecture A (mean moder-
 57 ation), the four families produced powers in $\{0.761, 0.769, 0.771, 0.778\}$, a 0.017
 58 spread within 2 MCSE of sampling noise. Under Architecture B (MVN differen-
 59 tial correlation), powers clustered at $\{0.779, 0.780, 0.781, 0.806\}$; logistic at 0.806
 60 sat $\approx$ 2 MCSE above the others, marginal and not reproduced under Architecture
 61 A. The piecewise-linear breakpoint family, hypothesised to behave qualitatively
 62 differently from the smooth families, was indistinguishable from gompertz and
 63 hyperbolic-tangent at this resolution under either architecture. Type I error at
 64 the null sat in $[0.025, 0.036]$ across all eight null cells, slightly conservative and
 65 without family-specific inflation.

66 **Conclusions.** The modified Gompertz family is a defensible default for the
 67 response-curve template in pmsimstats N-of-1 simulation, with explicit caveats for
 68 the trajectory regimes it cannot represent. We recommend reporting Gompertz-

69 vs-alternative sensitivity at least once per parameter regime where simulation
70 results inform protocol-design decisions, and we propose a default sensitivity grid
71 that the framework can run alongside its primary cells.

72 2 Introduction

73 2.1 A claim about response-curve choice in N-of-1 trial simulation

74 [1]

75 **The Gompertz function is a defensible default for the**
76 **saturating-shape regime in N-of-1 simulation.**

- 77 • Its asymmetric three-parameter form spans profiles that logistic
78 and exponential families cannot represent.
- 79 • Established precedent in tumour dynamics, microbial growth,
80 and the unified-Richards family supports the choice.
- 81 • The claim is conditional on the saturating-shape regime charac-
82 teristic of most pharmacodynamic trajectories over typical trial
83 timescales.

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86 The first claim is that the modified Gompertz function used as the
87 parametric template for treatment-response trajectories in aggregated
88 N-of-1 trial simulation (specifically, in the pmsimstats simulation
89 framework that operationalises the Hendrickson et al. design family¹)
90 is a defensible default for the saturating-shape regime that charac-
91 terises most pharmacodynamic and chronic-condition response curves
92 over the typical trial timescale. The function is well-established in
93 tumour growth dynamics^{2,3}, microbial growth modelling⁴, and the
94 broader unified- Richards growth-curve family⁵⁻⁷; its three-parameter
95 form is flexible enough to capture the asymmetric saturation profiles
96 that linear extrapolation, exponential decay, or symmetric logistic
97 models cannot.

98 [2]

99 **Family-misspecification sensitivity is absent from the N-of-1**
100 **literature and should be reported routinely.**

- 101 • Existing N-of-1 simulation studies inherit trajectory families with-
102 out comparing alternatives.
- 103 • Oncology and pharmacology modelling have established compar-
104 ative apparatus that the N-of-1 field has not imported.
- 105 • Routine sensitivity reporting is needed rather than silent assump-
106 tion of family adequacy.

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109 The second claim is that family-misspecification sensitivity is essen-
110 tially absent from the published N-of-1 methodology literature, and
111 that it should be reported routinely rather than assumed away. The
112 methodological N-of-1 corpus^{1,8–10} is, to the best of our knowledge,
113 silent on parametric trajectory shape: papers either inherit a chosen
114 family from the upstream literature without comparing alternatives,
115 or specify a linear or AR(1) form that is too coarse to capture the
116 saturation that real trajectories exhibit. The oncology-modelling lit-
117 erature has set the precedent for what a proper family-comparison
118 sensitivity analysis looks like^{6,11}, and the pharmacology literature has
119 converged on Hill / sigmoid-Emax / indirect-response models for re-
120 lated questions^{12,13}. The N-of-1 literature has not imported this com-
121 parative apparatus, and the present paper endeavors to supply it.

122 [3]

123 **The case for Gompertz is incomplete without a direct com-**
124 **parison of inferential consequences across alternative fami-**
125 **lies.**

- 126 • The paper does not assert Gompertz is uniquely correct but ar-
127gues it is reasonable within the unified-Richards class.
- 128 • Inferential consequences (power, type I error, bias) under alter-
129native families must be quantified, not asserted.
- 130 • The simulation study uses the standard pmsimstats linear-mixed
131analysis at trial-relevant sample sizes.

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134 The two claims are coupled because the case for a chosen family is
135 incomplete without the comparison. We do not argue that Gompertz
136 is the correct family in any strict sense; rather, we argue that it is
137 a reasonable choice within the unified-Richards family of saturating
138 curves, and that the inferential consequences of choosing it (versus the
139 symmetric logistic, the von Bertalanffy growth curve¹⁴, or a piecewise-
140 linear breakpoint model) should be quantified rather than asserted.
141 The paper reports a simulation study that does so for the standard
142 pmsimstats linear-mixed analysis at trial-relevant sample sizes.

143 2.2 Why the question matters

144 [4]

145 **Family choice is a DGP-level decision whose inferential cost**
146 **cannot be detected without varying the DGP family while**
147 **holding the analysis fixed.**

- A single-family simulation cannot reveal the inferential cost of that family choice.
- Robustness to family is a joint property of the DGP and analysis specification.
- This is analysis-model misspecification logic applied one level down in the simulation hierarchy.

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First, the chosen family loads into the data before any analysis-side question is asked. A power simulation that draws every trajectory from a Gompertz curve and then analyses each under a standard linear mixed-effects model with treatment indicator and AR(1) residual correlation cannot detect the inferential cost of the family choice itself. Robustness of the inference to the family is a property of the joint DGP-and-analysis specification; only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the same logic that underpins analysis-model misspecification studies in the broader clinical-trial methodology literature¹⁵, applied one level down in the simulation hierarchy, and it appears to us that the N-of-1 simulation literature has not yet drawn the parallel explicitly.

[5]

The Gompertz parameterisation has known identifiability fragility in short noisy trajectories that has not been characterised for N-of-1 designs.

- Asymptote and rate parameters correlate strongly; the shape parameter is near-unidentifiable with few inflection- region timepoints.
- Unified-Richards reparameterisations and sloppiness diagnostics exist but have not been applied to N-of-1 trajectory data.
- Comparative identifiability results from tumour-growth modelling support the need for an analogous N-of-1 diagnosis.

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Second, the function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.⁶ and⁷ propose unified-Richards reparameterisations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote)

with lower correlation, and¹⁶ develops the formal ‘sloppiness’ diagnostic that quantifies this directly as a property of the parameter space.¹¹ fitted eight tumour-growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalised logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories, and its absence is, in our view, problematic.

[6]

The Gompertz default in N-of-1 simulation reflects historical borrowing from growth biology, not empirical comparison with pharmacology alternatives.

- The pmsimstats Gompertz form traces to tumour dynamics and microbial growth literature, not pharmacodynamic evidence.
- The pharmacology literature defaults to Hill/sigmoid-Emax and indirect-response models for comparable questions.
- The two traditions have not been compared; inheritance is not itself evidence of adequacy.

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Third, the default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth-curve tradition, where its use traces to Laird’s tumour dynamics work² and to Zwietering’s microbial growth reformulations⁴. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family^{12,17} and to Jusko’s indirect-response models¹³. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defence against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence, and the question merits a direct empirical answer.

2.3 What the published N-of-1 literature does and does not do

[7]

No published N-of-1 simulation study has examined parametric trajectory-family choice or reported family-misspecification sensitivity.

- CONSORT-N-of-1, foundational programmatic statements, and

231 the Hendrickson framework all omit trajectory-family compari-
232 son.

- 233 • Silent inheritance of the Gompertz form is pervasive across ap-
234 plied N-of-1 simulation work.
- 235 • Adjacent literatures (oncology, pharmacometrics, latent- class
236 growth) have comparative apparatus that has not been applied
237 to N-of-1 simulation.

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240 A review of the recent N-of-1 methodology literature confirms, to the
241 best of our knowledge, a silence on the parametric-trajectory-family
242 question.

243 [8]

244 **Each major N-of-1 methodological strand silently inherits the**
245 **Gompertz default without family comparison.**

- 246 • CONSORT-N-of-1 reporting standards, Lillie and Schork pro-
247 grammatic statements, and the Hendrickson framework all omit
248 family comparison.
- 249 • Applied prazosin-PTSD work inherits the framework without re-
250 visiting the trajectory assumption.
- 251 • No published N-of-1 simulation study, to the authors' knowledge,
252 has broached the misspecification question.

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255 The CONSORT extension for N-of-1 trials¹⁰ codifies reporting stan-
256 dards but does not address parametric trajectory shape. The foun-
257 dational programmatic statements^{8,9} emphasise design and analysis
258 broadly without committing to a parametric response form. The Hen-
259 drickson et al. framework¹ introduces the modified Gompertz tem-
260 plate into the simulation engine for biomarker-validation power analy-
261 ses without comparing it against alternatives. Recent applied papers
262 in chronic-condition N-of-1 trials (including the prazosin-PTSD refer-
263 ence work building on¹⁸) inherit the framework without examining the
264 choice. Unfortunately, this pattern of silent inheritance is pervasive
265 enough that no published N-of-1 simulation study, to the best of our
266 knowledge, has broached the family-misspecification question.

267 [9]

268 **Comparative trajectory-family apparatus exists in adjacent**
269 **literatures but has not been applied to N-of-1 simulation.**

- 270 • Oncology modelling, pharmacometrics, and latent-class growth

literature each have established multi-family comparison methods.

- None of these methods has been applied with the Gompertz form as the N-of-1 simulation reference.
- Importing this machinery is the explicit contribution of the present paper.

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In adjacent literatures the comparative apparatus does exist. The oncology-modelling community routinely fits multiple growth families and compares them on identifiability and predictive performance¹¹. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models^{19,20}. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modelling²¹. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

[10]

The paper’s contribution is a four-family comparison against the standard pmsimstats analysis, reported under the ADEMP framework.

- Four families are compared: Gompertz, symmetric logistic, hyperbolic-tangent, and piecewise-linear breakpoint.
- The analysis model is held fixed at the standard pmsimstats linear-mixed specification.
- Results are reported in the Morris-White-Crowther ADEMP framework adopted as the project-wide standard.

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The present paper’s contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in the Morris-White-Crowther¹⁵ ADEMP framework already adopted as the project-wide simulation-reporting standard.

2.4 What this paper contributes

[11]

Section 2 characterises the pmsimstats Gompertz parameterisation and its relationship to the unified-Richards family.

- The vertical-offset adjustment and calibration parameter sets are documented.
- The unified-Richards reparameterisation provides more interpretable, less correlated parameters.
- This section establishes the reference point for all subsequent sensitivity comparisons.

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In section 2 we characterise the modified Gompertz family as implemented in pmsimstats: the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterisation^{6,7} that supplies more interpretable and less correlated parameters.

[12]

Section 3 surveys three alternative families at matched marginal effect size for visual and structural comparison with Gompertz.

- Families covered: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Each is calibrated to match the Gompertz reference at the same asymptote and anchor-point response level.
- Shape differences (symmetry, curvature, smoothness) are the primary axes of contrast.

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In section 3 we survey three alternative response-curve families: the symmetric logistic²², the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

[13]

Sections 4 and 5 report a Monte Carlo sensitivity simulation comparing power, type I error, and bias across the four trajectory families.

- Each family generates trajectories; all are analysed under the fixed standard pmsimstats linear-mixed model.
- Performance measures are reported with conventional Monte Carlo standard errors.

- 349 • The design borrows the Benzekry eight-model comparative
350 methodology and the Jagadeesan sloppiness diagnostic.

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353 In sections 4 and 5 we conduct a Monte Carlo sensitivity-analysis
354 simulation that generates trajectories under each of the four families
355 and analyses them under the standard pmsimstats linear-mixed model.
356 We report power, type I error, and bias of the biomarker-treatment
357 interaction estimate under each cell, with conventional Monte Carlo
358 standard errors. The simulation borrows the comparative methodol-
359 ogy of¹¹ (eight-model tumour-growth comparison) and the sloppiness
360 diagnostic of¹⁶ for small-sample identifiability assessment.

361 [14]

362 **Section 6 identifies the trajectory regimes where Gompertz**
363 **misspecification would be inferentially consequential and rec-**
364 **ommends sensitivity-grid extensions.**

- 365 • Consequential regimes include non-saturating growth, biphasic
366 patterns, and breakpoint behaviour.
- 367 • Standard N-of-1 inference is materially affected by family choice
368 in these regimes.
- 369 • Specific sensitivity-grid extensions are recommended for pmsim-
370 stats users.

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373 In section 6 we identify the trajectory regimes in which Gompertz mis-
374 specification would be expected to be consequential: non-saturating
375 growth beyond the trial window, biphasic patterns, and breakpoint be-
376 haviour. For these regimes, the standard N-of-1 inference is materially
377 affected by the family choice, and we recommend specific sensitivity-
378 grid extensions that pmsimstats users should run alongside their pri-
379 mary cells.

380 [15]

381 **A companion technical document preserves full mathematical**
382 **and historical context omitted from the manuscript proper.**

- 383 • Location: docs/25-gompertz-model-evaluation.md.
- 384 • Contains citations, derivations, and historical context that did
385 not survive manuscript editing.
- 386 • The manuscript is self-contained; the companion document is sup-
387 plementary reference material.

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390 The full mathematical and historical context, including citations that
391 did not survive editing into the manuscript proper, is preserved in the
392 companion technical document at docs/25-gompertz-model-evaluation.md.

393 3 The modified Gompertz function in pmsimstats

394 [16]

395 **The pmsimstats framework adopts a modified monotone- sat-**
396 **urating Gompertz function for each of three latent response**
397 **components.**

- 398 • Provenance traces to Gompertz’s 1825 mortality law and Win-
399 sor’s 1932 growth-curve application.
- 400 • Three components are parameterised: biological response BR,
401 placebo-belief response PB, and natural-history TV.
- 402 • The modified form is monotone-saturating, distinguishing it from
403 the original mortality-curve application.

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406 The Gompertz curve originates in Gompertz’s 1825 law of human
407 mortality²³ and was first applied as a general growth curve by²⁴; the
408 pmsimstats framework adopts the modified, monotone-saturating form
409 of that curve. The framework parameterises each of the three latent
410 response components (biological response BR, placebo-belief response
411 PB, and time-variant natural-history component TV) with a modified
412 Gompertz function

$$y(t) = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)), \quad (1)$$

413 [17]

414 **The three-parameter Gompertz form is asymmetric and**
415 **monotone-saturating, with the inflection location determined**
416 **jointly by rate and displacement.**

- 417 • Parameters are maxr (asymptote), rate (approach speed), and
418 disp (curvature near $t = 0$); a vertical offset ensures $y(0) = 0$.
- 419 • Inflection at $t^* = \log(\text{disp})/\text{rate}$; the asymmetry is the key struc-
420 tural distinction from logistic.
- 421 • Prazosin/PTSD reference values are $\text{maxr}_{BR} \approx 4$, $\text{rate}_{BR} \approx 0.5$,
422 $\text{disp}_{BR} \approx 4$.

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with vertical-offset adjustment so that $y(0) = 0$. The three parameters are `maxr` (asymptotic maximum response, matching the on-drug or on-natural-history saturation level), `rate` (which controls how quickly the response approaches its asymptote), and `disp` (which controls the curvature near $t = 0$). The Gompertz form is monotone-saturating and asymmetric: the inflection occurs at $t^* = \log(\text{disp})/\text{rate}$, with the rise being relatively slower before the inflection and faster after. The parameterisation used at the prazosin/PTSD reference cell sets $\text{maxr}_{BR} \approx 4$ nightmares per week, $\text{rate}_{BR} \approx 0.5$ per week, and $\text{disp}_{BR} \approx 4$ for biological response; analogous parameter sets for PB and TV are documented in `docs/25-gompertz-model-evaluation.md` and `R/utilities.R`.

4 Alternative response-curve families

[18]

Three alternative families are calibrated to match Gompertz at a common anchor point, isolating shape effects from amplitude effects.

- Families compared: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Calibration constraint: matched asymptote and response level at $t = 5$ weeks via `uniroot` or closed form.
- Families differ in smoothness, inflection symmetry, and curvature while sharing the same marginal effect size.

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Three alternative families serve as sensitivity-analysis comparators, and in this section we shall describe each in turn before the simulation study proper. Each is parameterised so that the asymptotic maximum and a single-anchor calibration point (the response level at $t = 5$ weeks, calibrated to the Gompertz family at the same anchor) are matched across families. That is, the families differ in shape envelope (smoothness, symmetry around the inflection, and curvature) while sharing the same marginal effect size at the anchor, a design choice that we discuss further in section 4.

4.1 Symmetric logistic

The logistic function

$$y(t) = \frac{\text{maxr}}{1 + \exp(-\text{rate} \cdot (t - t_0))} - \text{offset} \quad (2)$$

[19]

The symmetric logistic rises faster than Gompertz before the inflection and slower after, with symmetric inflection as the principal structural contrast.

- Symmetric around inflection at t_0 ; a vertical offset ensures $y(0) = 0$.
- At matched asymptote and anchor, the logistic envelope differs visibly from Gompertz in pre- and post-inflection curvature.
- Inflection symmetry is the primary qualitative contrast with the asymmetric Gompertz.

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is monotone-saturating and symmetric around its inflection at $t = t_0$. A vertical offset is subtracted so that $y(0) = 0$. At matched maxr and matched response level at the calibration anchor, the logistic family rises faster than Gompertz before the inflection and slower after, producing a visibly different shape envelope while sharing the same asymptote. The symmetric inflection is the principal qualitative contrast with the asymmetric Gompertz curve, and the reader will note that this symmetry property is, in a sense, the logistic family's most theoretically interesting feature for the purposes of the present comparison.

4.2 Hyperbolic-tangent

The hyperbolic-tangent variant

$$y(t) = \text{maxr} \cdot \frac{\tanh(\text{rate} \cdot (t - t_0)) + 1}{2} - \text{offset} \quad (3)$$

[20]

The hyperbolic-tangent family represents the fast- saturating smooth regime that neither the logistic nor Gompertz can span with standard parameterisations.

- Symmetric and monotone-saturating, but with a steeper pre-asymptote rise than the logistic.
- Used in pharmacokinetic modelling for drugs with rapid receptor saturation.
- Its inclusion spans the fast-saturating smooth regime absent from the other three families.

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is also monotone-saturating and symmetric, but with a qualitatively faster approach to the asymptote than the logistic. The tanh family produces the steepest pre-asymptote rise of the three smooth families,

and it has been used in pharmacokinetic modelling for drugs with rapid receptor saturation. We include it here as the representative of the fast-saturating smooth regime that neither the logistic nor the Gompertz family can span with their standard parameterisations.

4.3 Piecewise-linear breakpoint

The piecewise-linear-breakpoint family, formalised in the broken-line regression literature²⁵,

$$y(t) = \text{maxr} \cdot \min(1, t/t_{\text{bp}}) + \text{post_slope} \cdot \max(0, t - t_{\text{bp}}) \quad (4)$$

[21]

The piecewise-linear breakpoint is the canonical non- smooth alternative and provides the strongest test of the framework’s sensitivity to trajectory non-smoothness.

- A linear ramp to breakpoint t_{bp} then constant or small post-breakpoint slope.
- The breakpoint is the sole non-differentiable feature across the four families.
- Whether the Wald test detects the non-smooth feature is the central empirical question the simulation addresses.

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is the canonical non-smooth alternative: a linear ramp up to a breakpoint t_{bp} , then constant (when $\text{post_slope} = 0$) or with a small post-breakpoint slope. The breakpoint is the only non-differentiable feature across the four families and is the principal qualitative contrast with the three smooth families. This family is the strongest available test of the framework’s sensitivity to non-smoothness in the trajectory. But how different is the piecewise-linear breakpoint from the smooth families, from the perspective of the Wald test that the analysis model applies? That question is precisely what the simulation in section 4 is designed to answer.

5 Sensitivity-analysis simulation design

[22]

The primary estimand is the `bm:Dbc` interaction coefficient; performance measures are power, type I error, bias, and empirical Monte Carlo SE.

- Analysis model: $Sx \sim bm + t + Dbc + bm:Dbc + (1 \mid \text{ptID})$ with `corCAR1(form = ~t|ptID)`.

- Primary outputs: power at $\alpha = 0.05$ and type I error under $\beta_{bm:D} = 0$.
- Secondary outputs: bias and empirical MC SE of $\hat{\beta}_{bm:D}$; all proportions reported with binomial MCSE.

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The primary estimand is the biomarker-by-treatment interaction coefficient $\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model $Sx \sim bm + t + Dbc + bm:Dbc + (1 | ptID)$ with `corCAR1(form = ~t|ptID)`. Primary inferential outputs are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. Secondary estimands are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across replicates. Performance measures are reported with binomial-proportion Monte Carlo standard errors for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous quantities.

[23]

The ADEMP design is pre-registered and fixes all four DGP families, the calibration sub-study, and the full factorial before production runs.

- Pre-registration location: `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration`
- Study 0 calibrates all four families to a common marginal effect size; Study 1 is the 144-cell factorial; Study 2 is the sloppiness eigenvalue diagnostic.
- Pre-registration prevents post-hoc specification of performance measures or DGP choices.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pr` which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study (Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

[24]

The present submission uses a focused 16-cell factorial scoped to answer whether trajectory family matters for the bm:Dbc Wald test at MCSE 0.013–0.016.

- Design: 4 families $\times$ 2 architectures $\times$ 2 effect sizes; prazosin-calibrated OL+BDC, $N = 35$, $t_{1/2} = 0$.
- Carryover is absent to isolate trajectory shape from carryover-mediated correlation decay.
- The 144-cell extended grid is deferred to a follow-up; the 16-cell factorial addresses the central question at adequate MCSE resolution.

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For the present submission the simulation is scoped to a 16-cell focused factorial: four trajectory families crossed with two DGP architectures (mean moderation and MVN differential correlation, following the companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/`) crossed with two biomarker effect-size levels ($c_{bm} \in \{0, 0.45\}$). The trial design is the prazosin-calibrated Hybrid OL+BDC, $N = 35$, $t_{1/2} = 0$, with carryover set to absent in order to isolate the trajectory-shape effect from carryover-mediated correlation decay. Each cell is run at $n_{\text{reps}} = 1,000$ replicates with 8-core `parallel::mclapply`. The full 144-cell grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is deferred to a follow-up; the focused 16-cell factorial answers the central scientific question (does trajectory family choice matter for the `bm:Dgc` Wald test?) at MCSE 0.013–0.016 per cell.

[25]

All four families are matched at $t = 5$ weeks with the asymptotic maximum held constant; calibrated parameters are recorded for reproducibility.

- Anchor calibration: each family’s response at $t = 5$ matched to Gompertz via `uniroot` or closed form.
- Asymptotic maximum held constant across all four families.
- Calibrated parameters recorded in `01-architecture-a-trajectory-sweep.R` and read back by manuscript chunks.

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Calibration. The four trajectory families are matched at a single anchor point ($t = 5$ weeks): each family’s response level at $t = 5$ is set equal to the Gompertz reference level at $t = 5$ via `uniroot` (logistic, hyperbolic-tangent) or closed form (piecewise-linear breakpoint). The asymptotic maximum is held constant across families. Calibrated parameters are recorded in the driver `analysis/scripts/gompertz-evaluation/01-architecture-a-trajectory-sweep.R` and read back by the manuscript chunks for transparency.

6 Results

[26]

The 16-cell simulation completed with 100% convergence in 381 s; cell MCSE on power runs from 0.013 to 0.016.

- 1,000 replicates per cell across 16 cells; 8-core `parallel::mclapply`.
- Per-replicate output at `analysis/data/quick-sim/07-gompertz-replicates.rds`; ADEMP summary at `07-gompertz-summary.txt`.
- Results are presented in order: power, type I error, bias.

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The simulation programme was executed at 1{,}000 replicates per cell across the 16-cell trajectory-family-by-architecture-by-effect-size factorial in 381 s using 8-core `parallel::mclapply`. Convergence was 100% across all 16{,}000 fits. Per-replicate output is at `analysis/data/quick-sim/07-gompertz-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/07-gompertz-summary.txt`. Cell MCSE on power runs from 0.013 to 0.016 across the alternative cells and from 0.005 to 0.006 on the conservative null cells. We begin the results section with power, then turn to type I error, and finally consider bias of the interaction estimate.

6.1 Power across response-curve families

Trajectory-family power at $c_{bm} = 0.45$ under both architectures, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	76.1 $\pm$ 1.4	78.1 $\pm$ 1.3
Logistic	77.1 $\pm$ 1.3	80.6 $\pm$ 1.3
Hyperbolic tangent	76.9 $\pm$ 1.3	78.0 $\pm$ 1.3
Piecewise-linear breakpoint	77.8 $\pm$ 1.3	77.9 $\pm$ 1.3

[27]

Power is essentially invariant to trajectory family under both architectures; the pre-registered piecewise-linear hypothesis is not supported.

- Architecture A: four families within a 0.017 spread, well inside two cell-MCSE.
- Architecture B: spread 0.027; logistic at 0.806 is marginal (≈ 2 MCSE above others) and not reproduced under Architecture A.
- Piecewise-linear breakpoint is indistinguishable from Gompertz and hyperbolic-tangent under both architectures.

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652 We note that under Architecture A (mean moderation), the four fam-
653 ilies cluster within a 0.017 spread (1.7 percentage points), well inside
654 two cell-MCSE of sampling noise. Under Architecture B (MVN dif-
655 ferential correlation), the spread is 0.027; the logistic family at 0.806
656 sits roughly two MCSE above the other three families, which we read
657 as marginal at this resolution and not reproduced under Architecture
658 A. The pre-registered hypothesis that the piecewise-linear breakpoint
659 family would behave qualitatively differently from the smooth families
660 is not supported: at MCSE 0.013, the breakpoint family is indistin-
661 guishable from Gompertz and hyperbolic-tangent under Architecture
662 A, and again within two MCSE of them under Architecture B.

663 [28]

664 **The earlier 200-replicate prototype result was statistically**
665 **consistent with the production run but was under-powered**
666 **to distinguish the apparent gap from noise.**

- 667 • Prototype: Gompertz 0.765, logistic 0.805 under Architecture B;
668 apparent gap 0.040, roughly 1.5 prototype-MCSE.
- 669 • Production: gap settles to 0.025, within two production-MCSE
670 of zero.
- 671 • The prototype was not wrong; it lacked power to distinguish sig-
672 nal from noise at its replicate count.

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675 The 200-replicate prototype reported in the earlier draft of this
676 manuscript (Gompertz 0.765, logistic 0.805 under Architecture B
677 alone) is statistically consistent with the present production-replicate
678 results: the prototype’s apparent 0.040 Gompertz-logistic gap was
679 roughly 1.5 prototype-MCSE, and at the 1{,}000-replicate resolution
680 it settles to 0.025, itself within two production-MCSE of zero. That
681 is, the prototype result was not wrong, but it was under-powered to
682 distinguish signal from noise at that resolution.

683 6.2 Type I error across response-curve families

684 Type-I error at $c_{bm} = 0$, percent $\pm$ MCSE:

Family	Architecture A	Architecture B
Gompertz	2.6 ± 0.5	2.7 ± 0.5
Logistic	3.2 ± 0.6	2.9 ± 0.5
Hyperbolic tangent	2.9 ± 0.5	2.5 ± 0.5
Piecewise-linear breakpoint	2.8 ± 0.5	3.6 ± 0.6

[29]

Type I error is slightly conservative and uniform across all four families, consistent with corCAR1 small-sample degree-of-freedom accounting.

- All eight null cells fall in $[0.025, 0.036]$; no family-specific inflation is observed.
- The conservatism is shared across families, implicating the corCAR1 d.f. mechanism rather than trajectory shape.
- This pattern is consistent with prior pmsimstats runs at reference parameters and $N = 35$.

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All eight null cells produce empirical type I error in $[0.025, 0.036]$, slightly conservative relative to the nominal 0.05 and consistent with the corCAR1 small-sample degree-of-freedom accounting at the prazosin-calibrated $N = 35$ and 8-timepoint short serial series. There is no family-specific inflation. The conservatism is shared across all four families, supporting the interpretation that it reflects the corCAR1 d.f. mechanism rather than a trajectory-shape artefact. We note that this pattern of slight conservatism has appeared consistently across pmsimstats simulation runs at the reference parameters, and it is relatively well-understood as a consequence of the corCAR1 variance-component estimation at small N .

6.3 Bias of the biomarker-treatment interaction estimate

[30]

Bias of the interaction estimate is negligible and uniform across families; within-replicate model SE matches empirical SE across all cells.

- Mean $\hat{\beta}_{bm:D_{bc}}$ varies from -0.230 to -0.239 across the eight alternative cells, a 0.009 spread within Monte Carlo noise.
- Empirical SE is essentially constant across families (0.074 to 0.082).
- Model SE matches empirical SE, confirming well-calibrated nominal-95% Wald coverage.

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Across the eight alternative cells the mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ varies over a narrow range (roughly -0.230 to -0.239), a cell-to-cell spread of about 0.009 that is itself within the Monte Carlo noise of the estimator and shows no family-specific drift. The empirical

SE of $\hat{\beta}_{bm:D_{bc}}$ is essentially constant across families (SD in the range 0.074 to 0.082 across cells) and roughly equal between architectures. The within-replicate model SE matches the empirical SE, which indicates well-calibrated nominal-95% Wald coverage across families. In sum, the bias results are unremarkable in the best sense: there is no trajectory-family signal here that the power and type I error results might have concealed.

7 Discussion

7.1 A negative finding, cleanly resolved

[31]

Trajectory family choice has no detectable effect on the bm:D_{bc} Wald test at the prazosin reference cell under either DGP architecture.

- Four families spanning asymmetric, symmetric-smooth, and non-smooth templates all produce statistically indistinguishable power and type I error.
- The finding is a null result at one reference cell and should not be overstated.
- Conditions under which the result might not hold are discussed in section 6.3.

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We report what is, in essence, a negative finding: at the prazosin-calibrated reference cell (Hybrid OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 0$, matched-anchor calibration), trajectory family choice has no detectable effect on the bm:D_{bc} Wald test under either DGP architecture. The four families span the qualitative range of plausible templates for monotone-saturating treatment-response trajectories: asymmetric (Gompertz), symmetric smooth (logistic and hyperbolic-tangent), and non-smooth (piecewise-linear breakpoint). All four produce statistically indistinguishable power and type I error at MCSE 0.013 to 0.016 per cell. We would not wish to overstate this finding; it is a null result at one reference cell, and the conditions under which the result might not hold are discussed in section 6.3 below.

[32]

The pre-registered piecewise-linear hypothesis is not supported; the breakpoint feature is invisible to the Wald test in the explored regime.

- The non-differentiable breakpoint carries no detectable power

764 cost at $N = 35$ and $c_{bm} = 0.45$.

- 765 • Smoothness assumptions in the trajectory family carry no infer-
766 ential cost in the regime evaluated.
- 767 • This null result is methodologically informative: it implies the
768 Wald test is robust to the non-smooth feature.

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771 The pre-registered hypothesis that the piecewise-linear breakpoint
772 family, with its non-differentiable feature, would behave qualitatively
773 differently from the smooth families is not supported. The breakpoint
774 feature is invisible to the linear-mixed Wald test at the explored
775 sample size and effect size. This is a methodologically valuable
776 null result: if the breakpoint feature were detectable through power
777 loss, the manuscript would have reported a meaningful sensitivity to
778 non-smoothness. Its absence implies that smoothness assumptions in
779 the trajectory family carry no inferential cost in the regime evaluated.

780 7.2 Why trajectory family does not matter at this resolution

781 [33]

782 **Three mechanisms jointly account for the trajectory- family**
783 **null finding.**

- 784 • Calibration absorbs shape differences at the common anchor, iso-
785 lating shape from amplitude effects.
- 786 • The bm:Dbc Wald test integrates over time, averaging local shape
787 differences within the predictor structure.
- 788 • Under Architecture B, biomarker-treatment information lives in
789 the covariance structure, upstream of and independent from tra-
790 jectory shape.

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793 We see three mechanisms that, taken together, account for the nega-
794 tive finding, and we shall work through each in turn.

795 [34]

796 **Matched-anchor calibration ensures that residual family dif-**
797 **ferences are in shape only, not in amplitude, and those shape**
798 **differences are small relative to between-replicate variation.**

- 799 • Families matched at asymptote and $t = 5$ anchor.
- 800 • Within-trajectory shape variation is small relative to the between-
801 replicate variance of the bm:Dbc estimator.

- The calibration design isolates shape effects from amplitude effects by construction.

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The first mechanism is that calibration absorbs the differences. The four families are matched at the asymptotic maximum and at a single anchor point ($t = 5$ weeks). At matched marginal effect size, the within-trajectory variation in shape is small relative to the between-replicate variation in the bm:Dbc estimator. The calibration constraint is intentional in the simulation design: it isolates trajectory-shape effects from amplitude effects, so that any family-specific power differences we observe can be attributed to shape alone.

[35]

The bm:Dbc Wald test averages over on-drug timepoints, smoothing out local trajectory-shape differences that would be visible in point-in-time comparisons.

- The continuous D_{bc} predictor integrates trajectory contributions across all on-drug timepoints.
- Local shape differences are averaged within the integration and do not accumulate to detectable power differences.
- This temporal averaging appears to be the dominant mechanism behind the near-identical power.

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The second mechanism is that the bm:Dbc Wald test averages over time. The continuous D_{bc} predictor multiplied by the biomarker integrates the trajectory's contribution across all on-drug timepoints, with weights determined by the predictor's exponential-decay structure. Local trajectory-shape differences (where one family rises faster than another between timepoints) are averaged within the integration, and the Wald test is therefore relatively insensitive to local-shape features. It appears that this averaging property is the dominant reason for the near-identical power across families.

[36]

Under Architecture B, the biomarker-treatment information channel is in the covariance structure and is independent of trajectory shape by construction.

- Architecture B encodes information in $\partial \text{Cor}(BM, BR) / \partial D_{bc}$, not in the trajectory mean.
- This channel is independent of the trajectory family that BR follows.

- 843 • Family choice cannot modulate Architecture B’s information by
844 construction; invariance is expected.

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847 The third mechanism applies specifically to Architecture B: the co-
848 variance moderation is upstream of trajectory shape. Under MVN
849 differential correlation, the biomarker-by-treatment information lives
850 in the $\partial\text{Cor}(BM, BR)/\partial D_{bc}$ structure, which is independent of the
851 trajectory family that BR follows. Family choice cannot affect Archi-
852 tecture B’s information channel by construction.

853 [37]

854 **Trajectory family is inferentially inert at this resolution; the**
855 **modGompertz default is robust by the combined action of**
856 **the three mechanisms.**

- 857 • Calibration, temporal averaging, and covariance-upstream encod-
858 ing jointly suppress family-specific power variation.
- 859 • The modGompertz default is robust at the reference parameters
860 under both architectures.
- 861 • The following section identifies conditions where this robustness
862 would be expected to break down.

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865 The combined effect is that trajectory family choice is, at this res-
866 olution, an inferentially inert axis of the simulation design, and the
867 framework’s modGompertz default is in this sense robust by construc-
868 tion. We turn next to the conditions under which that robustness
869 would be expected to break down.

870 7.3 Conditions that would change the answer

871 [38]

872 **Four conditions outside the present 16-cell factorial would**
873 **plausibly produce trajectory-family-dependent power.**

- 874 • Smaller N , different c_{bm} , non-zero carryover ($t_{1/2} > 0$), and non-
875 monotone or biphasic shape envelopes.
- 876 • None of these conditions is exercised in the present focused fac-
877 torial.
- 878 • Each is a natural axis for the 144-cell follow-up grid.

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Four conditions, none exercised in the present 16-cell factorial, would plausibly produce trajectory-family-dependent power, and each merits a brief comment.

[39]

Smaller sample size might amplify local-shape sensitivity by concentrating temporal information at fewer timepoints.

- At $N = 35$, the test integrates over a moderately large within-subject sample.
- With fewer timepoints, local shape differences between families would be more visible rather than averaged.
- This condition is the most proximate candidate for a departure from the null result.

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First, smaller sample size. At $N = 35$ the `bm:Dbc` test integrates over a moderately large within-subject sample; smaller N might amplify the local-shape sensitivity by concentrating the temporal information at fewer timepoints where the families differ more visibly.

[40]

At smaller c_{bm} near the 50% power boundary, family differences might surface via different finite-sample sampling distributions.

- The $c_{bm} = 0.45$ cell sits in a moderate-power regime with comparable signal-to-noise across families.
- Pushing to $c_{bm} = 0.30$ (power near 0.5) may reveal family-specific finite-sample sampling differences.
- The tight clustering at 0.45 may be a local feature of the effect-size space.

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Second, larger or smaller c_{bm} . The $c_{bm} = 0.45$ reference cell sits in a power regime where families have comparable signal-to-noise. Pushing c_{bm} to 0.30 (where power lies near 0.5) might surface family differences via different finite-sample sampling distributions, and it is possible that the tight clustering we observe is a feature of this particular region of the effect-size space.

[41]

Non-zero carryover introduces an interaction between trajectory shape and the off-drug exponential decay that is suppressed in the present $t_{1/2} = 0$ slice.

- With $t_{1/2} > 0$, integration weights interact with the off-drug decay, potentially differentiating families.
- The present simulation sets $t_{1/2} = 0$ to isolate trajectory shape from carryover effects.
- This interaction is the mechanistic pathway by which carryover could make trajectory family consequential.

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Third, carryover-induced correlation decay. With $t_{1/2} > 0$, the bm:Dbc-mediated information channel is eroded differently for different trajectory shapes, because the integration weights interact with the off-drug exponential decay. The $t_{1/2} = 0$ slice exercised here suppresses this interaction entirely.

[42]

Non-monotone or biphasic trajectory families would produce genuinely different bm:Dbc information content to which the Wald test would almost certainly respond.

- All four present families share a monotone-saturating envelope; the null result is conditional on this property.
- Biphasic or oscillating families have qualitatively different information structure.
- Extension to non-monotone families is outside the scope of the present study.

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Fourth, non-monotone or biphasic shape envelopes. The four families we have examined share a monotone-saturating envelope. Families that peak and decay (biphasic) or oscillate would produce genuinely different trajectories with different bm:Dbc information content, and the Wald test would almost certainly respond to those differences.

[43]

The 144-cell extended grid is the canonical follow-up that would close the trajectory-family question at publication-adequate rigour.

- The grid crosses N , c_{bm} , $t_{1/2}$, and family.
- Each of the four conditions above maps to a dimension of this grid.
- The present focused factorial is a necessary precursor; the extended grid is the production-grade completion.

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962 Each condition is the natural axis of a follow-up sensitivity analysis.
963 The 144-cell grid anticipated in the abstract ($\$N \times c_{\{bm\}} \times t_{\{1/2\}}$
964 $\times \$$ family) is the canonical follow-up, and we recommend it as the
965 production-grade extension that would close out the trajectory-family
966 question at a level of rigour adequate for the methodological literature.

967 7.4 The prazosin/PTSD reference cell

968 [44]

969 **For the prazosin-PTSD application, trajectory family is the**
970 **least consequential sensitivity axis; attention should be di-**
971 **rected to carryover and DGP-architecture.**

- 972 • Sensitivity to modGompertz is small relative to carryover speci-
973 fication and DGP-architecture sensitivity.
- 974 • The modGompertz choice can be defended without reservation
975 at the reference parameters.
- 976 • Sensitivity-analysis resources are better spent on carryover and
977 DGP-architecture axes with demonstrably larger inferential con-
978 sequences.

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981 For the motivating prazosin-PTSD application that drives the pmsim-
982 stats research programme, the present result implies that sensitivity
983 to the modGompertz assumption is small relative to sensitivity to the
984 carryover specification (the subject of the companion manuscript at
985 `analysis/report/02-carryover-sensitivity/`) and to the DGP ar-
986 chitecture (the subject of `analysis/report/01-dgp-mean-moderation-vs-mvn/`).
987 That is, of the three principal axes of sensitivity uncertainty in the
988 pmsimstats simulation framework, trajectory family choice is, at the
989 reference parameters, the least consequential by a substantial margin.
990 A trial designer using the pmsimstats framework can defend the
991 modGompertz choice without reservation at the reference parameters,
992 and we would suggest that sensitivity-analysis attention be redirected
993 to the carryover and DGP-architecture axes, where the inferential
994 consequences are demonstrably larger.

995 8 Conclusions

996 [45]

997 **The modified Gompertz is a defensible template for**

998 **monotone-saturating N-of-1 trajectories; four-family com-**
999 **parison produces indistinguishable power and type I error.**

- 1000 • Power and type I error are statistically indistinguishable across
- 1001 all four families under both architectures at MCSE 0.013–0.016.
- 1002 • The piecewise-linear breakpoint hypothesis is not supported.
- 1003 • The Gompertz default is defensible at the reference parameters.

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1006 First, the modified Gompertz function is a defensible parametric tem-
1007 plate for monotone-saturating treatment-response trajectories in N-
1008 of-1 simulation. At the prazosin/PTSD reference cell with matched-
1009 anchor calibration, the four-family comparison (Gompertz, symmet-
1010 ric logistic, hyperbolic-tangent, piecewise-linear breakpoint) produces
1011 statistically indistinguishable power and type I error under both DGP
1012 architectures at MCSE 0.013 to 0.016 per cell. The pre-registered
1013 hypothesis that the piecewise-linear breakpoint family would behave
1014 qualitatively differently from the smooth families is not supported.

1015 [46]

1016 **The null result implies the Wald test is robust to princi-**
1017 **pal trajectory-family axes in the evaluated regime; boundary**
1018 **conditions are deferred to the 144-cell follow-up.**

- 1019 • Robustness holds across asymmetric, symmetric-smooth, and
- 1020 non-smooth families in the evaluated regime.
- 1021 • At what sample sizes, effect sizes, or carryover levels robustness
- 1022 breaks down is an open question.
- 1023 • Sensitivity-analysis priority should go to carryover and DGP-
- 1024 architecture axes with larger demonstrated consequences.

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1027 Second, the negative finding is methodologically informative. It im-
1028 plies that the bm:Dcb Wald test is robust to the principal qualita-
1029 tive axes of trajectory-family choice in the regime evaluated. Unfortu-
1030 nately, we cannot yet establish at what sample sizes, effect sizes, or car-
1031 ryover levels that robustness would begin to break down; that question
1032 is deferred to the 144-cell follow-up. Trial designers using the pmsim-
1033 stats framework can accept the modGompertz default without reser-
1034 vation at the reference parameters and direct sensitivity-analysis at-
1035 tention to the carryover and DGP-architecture axes that have demon-
1036 strably larger inferential consequences.

1037 [47]

The 144-cell extended grid is the work needed to close the trajectory-family question at publication-adequate rigour.

- The present 16-cell factorial addresses the central question with statistical clarity.
- The extended grid would map boundary conditions at which trajectory family begins to matter.
- This follow-up is recommended as the production-grade completion of the trajectory-family evaluation.

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Third, the 144-cell extended grid (sample sizes, effect sizes, DGP-side carryover) anticipated in the original abstract is the natural follow-up, and we regard it as the work that would be needed to close the trajectory-family question at a level of rigour adequate for the methodological literature. The present focused 16-cell factorial addresses the central scientific question with statistical clarity; the extended grid would map the boundary conditions at which trajectory-family choice begins to matter.

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

[48]

Data and code are openly available in the pmsimstats-ng repository under versioned standard paths.

- Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are at `analysis/data/`.
- Simulation drivers are at `analysis/scripts/`.
- Exact manuscript paths are listed in the Reproducibility section.

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at

analysis/scripts/. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

[49]

The manuscript is fully reproducible inside the zzcollab Docker container with the renv.lock package set.

- Container image hash in Dockerfile; package set in renv.lock.
- Principal packages: nlme, parallel, data.table, furrr/purrr, rmarkdown/bookdown.
- Rendered with R 4.4.x.

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in Dockerfile; package set captured in renv.lock). Principal packages used by the simulation pipeline are nlme (continuous-time linear-mixed analysis with corCAR1 residual correlation), parallel (8-core mclapply for parameter-grid sweeps), data.table (the original-extended simulation collection), furrr and purrr (the tidyverse simulation collection), and rmarkdown plus bookdown for the manuscript build.

[50]

Seeds, checkpoints, and driver paths are fully specified for exact replication of the simulation results.

- Master seed `set.seed(20260507)`; per-replicate seeds via `parallel::mc.set.seed` with `+ rep_idx` offsets.
- Outputs checkpointed at `analysis/data/quick-sim/07-gompertz-replicates.rds`; ADEMP summaries at `07-gompertz-summary.txt`.
- Pre-registered ADEMP plan at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registr`

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/07-gompertz-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `07-gompertz-summary.txt`. Driver scripts are under `analysis/scripts/gompertz-evaluation/`. The pre-registered ADEMP plan is at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registr`

1112 The manuscript source, paper-specific bibliography (`references.bib`),
 1113 and SIM CSL file are at `analysis/report/07-gompertz-evaluation/`.

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